Introduction of Financial Professional Certificate



Equity Investments

Definition

One or more shares in the ownership of a business or corporation that are purchased by investors who are then entitled to shares of the firm's assets in the case of liquidation.

These shares of stock may be bought and sold among stockholders in response to changes in market price.

Dividend Discount Model (DDM)

Value of a Preferred Stock

$$= \frac{D_1}{(1+k)^1} + \frac{D_2}{(1+k)^2} + \ldots + \frac{D_\infty}{(1+k)^\infty} \; or \; \frac{D}{k}$$

(1) Unlike common equity, preferred stocks pay a fixed dividend.
(2) The value of a preferred stock can be calculated using the dividend discount model.
(3) The value of the preferred stock is essentially the present value of the dividend in perpetuity, where k is the required return.

Example: Calculating the value of a preferred stock

Assume that Tamkang's preferred stock pays out to an investor an annual dividend of $8 per share. Given a rate of return of 10%, what is the value of Tamkang's preferred stock?

Value of a Common Stock



- Much like a preferred stock, holders of common stock can also receive dividends.
- But, dividends on common stock are not guaranteed, nor are they a fixed amount from year-to-year.
- As such, we can value common stock using dividends over various time horizons.

1. One-year Holding period

Consider a one-year holding period, in this time frame, an investor will receive a dividend and the price of the common stock at the end of the one year, both discounted back by the rate of return on the common equity.

➔ we can calculate the value of the common stock as follows:

$$\text{Common Stock} = \frac{\text{Dividend}}{(1+k)^1} + \frac{\text{Price at the end of the year}}{(1+k)^1}$$

Example: Calculate the value of a stock with a 1-year holding period
Tamkang expects to pay its shareholders common equity, $0.25 per share this year. The investor anticipates Tamkang's stock will close the year at $30 per share. Given a rate of return of 10%, what is the value of Tamkang's common stock?

2. Multiple-year holding period

Given investors can hold a common stock for over a year, it is useful to value a stock over the investor's expected holding period. In this case, the DDM model can be used.

Value of common equity

$$= \frac{D_1}{(1+k)^1} + \frac{D_2}{(1+k)^2} + \ldots + \frac{D_n}{(1+k)^n}$$

Example: Calculate the value of a stock with a multiple-year holding period

An investor plans to hold Tamkang's stock for 2 years. Tamkang expects to pay its shareholders common equity, $0.25 per share over the next two years. The investor anticipates Tamkang's stock will close the end of that time period at $40 per share. Given a rate of return of 10%, what is the value of Tamkang's common stock at the end of the two-year time period?

3. Infinite Period DDM

To value a common stock using the infinite period DDM, the calculation is simplified much like it is when valuing a preferred stock with infinite dividends.

$$\text{Common stock} = \frac{D_1}{k - g}$$

Where k = rate of return,
g = growth rate

Example: Calculate the value of common stock with infinite dividends

Tamkang paid a $0.25 annual dividend in 2016 and expects to pay $0.265 in 2017. Assuming a 6% growth rate and a required rate of return of 10%, calculate the value of Tamkang's stock.

Valuing the Common Stock of Companies with Supernormal Growth

When valuing a common stock of a company which is experiencing significant growth, we take a similar approach to valuing the common stock with a multi-year holding period.

The difference in the approaches is related to the dividend. A multi-year holding period approach assumes a stable dividend, whereas the dividend changes given the supernormal growth in this approach.

$$\text{Value of common equity} = \frac{D_1}{(1+k)^1} + \frac{D_2}{(1+k)^2} + \ldots + \frac{P_n}{(1+k)^n}$$

Example: **Calculate the value of common stock with temporary supernormal growth**

An investor plans to hold Tamkang's stock for 3 years. In that time period, Tamkang plans to grow at a rate of 6% in the first two years and 3% thereafter. Tamkang's last dividend was $0.25. Given a rate of return of 10%, what is the value of Tamkang's common stock at the end of the three-year time period?

Using the DDM to Develop an Earnings Multiplier Model

Developing the earnings multiplier using the DDM begins with the basic determinant of price.

To determine the price of a stock using the DDM, the infinite dividend is divided by the required return minus the growth rate as follows:

$$P_0 = \frac{D_1}{(k-g)}$$

To determine the price to earnings multiple, the price of the stock is simply divided by the earnings per share of the stock as follows:

$$\text{Earnings Multiplier} = \frac{P_0}{EPS_1} = \frac{D_1/EPS_1}{(k-g)}$$

Example: Determining a company's price-to-earnings ratio using the DDM with Tamkang's $0.25 dividend payout, an EPS of $1.00, calculate the stock's P/E ratio assuming 10% required return and 5% growth.

The Components of An Investors' Required Rate of Return



Real risk-free rate -

- This rate assumes no inflation or risk is prevalent, but that it is simply generated by the supply and demand of the markets.

Expected rate of inflation -

- This rate anticipates the potential inflation that is going to occur in the market.

Risk premium -

- The premium is reflective of the risks inherent in the stock, as well as the market. Such risks include liquidity risk, business risk and general macroeconomic risk.

The Implied Dividend Growth Rate



- A company's dividend growth rate can be derived from a company's ROE and its retention rate.
- The retention rate of a company is the amount of earnings a company retains for its internal growth.
- A company's ROE is the return on the funds invested back into the company. Note: the growth rate of the firm is the identified ability of a firm to grow its operations and the ROE is the return the company is able to earn on invested funds.
- The company's growth rate can be calculated as follows:
  Growth rate = (retention rate)(ROE)

Example: Estimate a dividend growth rate given ROE and retention rate

Tamkang assumes a constant ROE of 15%. The company anticipates a retention rate of 60% to fund new projects (indicating the firm will pay out 40% in dividends). What is Tamkang's dividend growth rate?

To determine stock price



To determine a company's stock price using the DDM the following inputs must be available:

1. **Required Rate of Return**
This return can be derived from the CAPM model (Rcs = Rf + Bcs(Rmarket - Rf)

2. **Expected Growth Rate**
The expected growth rate can be derived from both the retention rate (1-payout rate) and the company's ROE. (g = (retention rate)(ROE))

To determine stock price



3. Dividend

- To determine the dividend to be used in the DDM, last year's dividend must first be calculated as the dividend in the DDM is next year's dividend.
- Given a company's earnings last year as well as the company's payout ratio, last year's dividend can be calculated.
- Using the growth rate derived above, next year's dividend can be determined.

4. Future Price

The future price for the company can be derived with next year's dividend divided by the difference in the company's required rate of return and its growth rate.

Example: Calculate the required inputs to be used in the DDM

Tamkang's annual dividend last year was $1.00. The company maintained its annual dividend payout ratio of 40% and ROE of 16%. Tamkang's beta is 1.3. Given a risk-free rate of 4% and an expected return on the market of 18%, determine Tamkang's required rate of return, expected growth rate and next year's dividend and price?

Earnings Per Share of a Stock Market Series

The earnings per share of a stock market series can be calculated as follows:

$$EPS_{\text{stock market series}} = [(\text{sales})(\text{OM}) - \text{depreciation} - \text{interest}] \times (1\text{-tax})$$

Where:

- Sales = sales per share of the estimated series is determined through regression
- OM = the operating margin is typically calculated as a percentage of sales
- Depreciation = this is determined by either continuing the trend of the current depreciation or focusing on the expected capital expenditures and how that number relates to future dividends.
- Interest = interest is determined by outstanding debt and the interest rate on that debt

Example: Calculate the EPS of a stock market series

Assuming the following estimates for a stock market series, sales per share = $100.00, OM = 50%, depreciation of $20.00, interest of $2.00 and a corporate tax rate of 40%, calculate the forecasted EPS of the stock market series?

The Expected Return of a Stock Market Series

Similar to determining the earnings multiplier of a stock, the earnings multiplier of a stock market series can be derived by the DDM.

Earnings multiplier

$$= \frac{P_0}{EPS_1} = \frac{D_1 / EPS_1}{(k - g)}$$

In addition to the EPS, the series' required rate of return, growth and dividend are needed.

Example: Calculate the expected P/E ratio of a stock market series

Next year's EPS for the series was determined to be $16.80. With a dividend of $4.00 for next year's dividend payout, calculate the stock series' earnings multiplier assuming 10% required return and 5% growth.

Estimate and Interpret the Earnings Multiplier of a Stock Market Series

To determine value of the earnings multiplier for a stock market series the following inputs must be determined:

1. EPS

Next year's EPS must be forecasted to determine the earnings multiplier.

2. Required Rate of Return

This return is derived from the CAPM model (Rseries = Rf + Bseries(Rmarket - Rf). If the required rate of return increases, the multiplier decreases.

3. Expected Growth Rate

The expected growth rate can be derived from both the retention rate (1-payout rate) and corporate ROE. (g = (retention rate)(ROE)). If the expected growth rate increases, the multiplier increases.

Estimate and Interpret the Earnings Multiplier of a Stock Market Series



4. Dividend

To determine the dividend to be used in the DDM, last year's dividend must first be calculated as the dividend in the DDM is next year's dividend. Given last year's corporate earnings as well as the corporate payout ratio, last year's dividend can be calculated. Using the growth rate derived above, next year's dividend can be determined.

5. Future price

The future price for the company can be derived with next year's dividend divided by the difference in the company's required rate of return and its growth rate.

The Expected Rate of Return for a Stock Market Series

The expected return rate of return of a stock market series, including the dividend, is the relative change that is anticipated in the stock market series over a specified time period.

$$\mathbf{E(R)} = \frac{\text{End value of series} + \text{Dividend} - \text{Beg. value of series}}{\text{Beg. value of series}}$$

Example: Calculate the expected rate of return for a stock market series

For a stock market series, it is estimated to have a dividend of $5.00 and an ending value of $25. If the current value of the stock market series is $15, calculate the expected return for the stock market series.

Company's Expected EPS



Similar to a stock market series, a company's expected earnings per share (EPS) can be determined as follows:

$$EPS_{\text{stock market series}} = [(\text{sales})(\text{operating margin}) - \text{depreciation} - \text{interest}] \times (1 - \text{tax})$$

Where:

- **Sales** = sales per share of the estimated series determined through regression
- **OM** = the operating margin is typically calculated as a percentage of sales
- **Depreciation** = this is determined by either continuing the trend of the current depreciation or focusing on the expected capital expenditures and how that number relates to future dividends.
- **Interest** = interest is determined by outstanding debt and the interest rate on that debt.

The inputs, however, are determined from specific company data.

Example: Calculate the EPS of a company

Assuming the following estimates for a company, sales per share = $50.00, OM = 20%, depreciation of $5.00, interest of $1.00 and a corporate tax rate of 40%, calculate the forecasted EPS of the company?

Determining the Earnings Multiplier of a Company

Similar to a stock market series, a company's earnings multiplier can be determined as follows:

The price of the stock is simply divided by the earnings per share of the stock as follows:

Earnings multiplier

$$= \frac{P_0}{EPS_1} = \frac{D_1 / EPS_1}{(k - g)}$$

Example: Determining a company's earnings multiplier

With Tamkang's $0.25 dividend payout, an EPS of $1.00, calculate the stock's P/E ratio assuming 10% required return and 5% growth.